

Kindred: abstract-based reviewer ranking for OR/MS papers

Technical note

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1 Purpose

Kindred proposes reviewers for a manuscript. The input is the manuscript’s title and abstract, typed by the user, or a PDF. The output is a ranked list of authors of recent OR/MS papers. Each author comes with the papers that support the match and with any potential conflict of interest and its evidence.

All computation on the manuscript runs on the user’s machine. The manuscript, its embedding and its reference list never leave the machine. The command `kindred verify-offline` runs the pipeline with network sockets disabled and fails if any network call is attempted. Typing the title and abstract, rather than uploading the PDF, means the tool never handles the full manuscript at all.

2 The two databases

Paper database. One row per paper: OpenAlex work ID, DOI, title, abstract, year, venue, open-access URL, citation count, and the list of referenced works. The source is OpenAlex (CC0). The corpus is every work published from 2014 onward in 29 OR/MS venues (Appendix A). It holds 74,127 papers. Abstracts are present for 62.5%; the gap is almost entirely Elsevier journals, whose abstracts OpenAlex does not carry. Semantic Scholar supplied 7,473 of the missing abstracts. The provenance and the pull procedure are in Appendix B.

Author database. One row per (paper, author) pair: paper ID, OpenAlex author ID, author name, author position, and the first-listed institution ID and name. There are 101,183 distinct authors. Author IDs are OpenAlex’s disambiguated identities, so a person appears once even when their name is written differently across papers. The reviewer pool is exactly this set of authors. A side table holds per-author statistics from OpenAlex (works count, citations, h -index), and further tables hold editorial-board membership and advisor/student links (Appendix C).

3 Embeddings

Let ϕ map a title and an abstract to a unit vector in $\mathbb{R}^{768}$:

$$e_i = \phi(\text{title}_i, \text{abstract}_i), \quad \|e_i\|_2 = 1.$$

ϕ is the SPECTER2 encoder: a SciBERT transformer (`allenai/specter2_base`) with the `proximity` adapter. The input string is the title, the separator token, and the abstract, truncated to 512 tokens. The output is the final-layer `[CLS]` vector, normalised to unit length. SPECTER2 was

trained so that papers which cite each other have nearby vectors. As a result the inner product $\langle e_i, e_j \rangle$ (the cosine similarity) is high for papers on the same topic and method, even when they share few words.

The index stores e_i for every paper with a title (73,643 vectors, float16, about 110 MB). The manuscript is mapped by the same function: $q = \phi(\text{title}, \text{abstract})$. Because query and index use the same ϕ , similarities are comparable. Embedding the full corpus takes about 200 s on one RTX 4090 GPU. On a 24-thread CPU the rate is about 0.08 s per paper, so the full corpus would take about 1.6 hours; a single query takes 0.09 s on CPU, so users without a GPU lose nothing at query time.

Who uses such embeddings. OpenReview’s expertise model, used for reviewer assignment at NeurIPS, ICLR and ICML, scores reviewer–paper pairs with SPECTER embeddings of titles and abstracts (mixed 0.8/0.2 with a second model), and pools a reviewer’s papers by the maximum cosine. Semantic Scholar uses SPECTER for its paper-recommendation features. In the gold-standard study of Stelmakh et al., which compared matching algorithms against self-reported reviewer expertise, SPECTER-family models with max pooling were the most accurate; TPMS, which uses word counts over full text, gained only a little from the full text. Kindred uses the same family of embeddings and the same pooling choices.

4 The ranking problem

Given the manuscript vector q , the task is to rank the authors a in the author database by how well they fit as reviewers, excluding or flagging those with a conflict of interest. Let P_a be the set of papers of author a and $s_i = \langle q, e_i \rangle$ the similarity of paper i to the manuscript. The fit of an author is a function of the similarities of their papers. The basic score is

$$\text{score}(a) = (1 - \lambda) \max_{i \in P_a} s_i + \lambda \text{mean}_{\text{top-}k} \{s_i : i \in P_a\},$$

that is, a mix of the author’s single most similar paper and the mean of their k most similar papers. With $\lambda = 0$ this is max pooling. Ranking authors by this score is the whole task: it is a nearest-neighbour search over papers, aggregated to authors. Kindred does not solve an assignment problem. There is no assignment or capacity constraint; the tool proposes a list and a person decides.

Further terms adjust the score. All are optional and all are in cosine units, so they are comparable with each other and with s_i . Appendix D gives the exact form of each.

- **Citation bonus.** Authors of papers the manuscript cites, taken from its reference list. This is the largest single improvement in the evaluation.
- **Volume correction.** Removes the advantage of prolific authors, who get more chances at a high $\max s_i$ by having more papers.
- **Editorial and seniority penalties.** Reduce the score of authors with current or past editorial roles, and of authors with many publications.
- **Early-career bonus.** Raises the score of authors with few publications, such as PhD students and postdocs.
- **Seed bonus.** Raises the score of authors whose body of work resembles seed reviewers named by the user.

- **Diversity re-ranking.** Re-orders the list so consecutive candidates come from different research neighbourhoods.

The interface displays the objective with the current weights and, for each candidate, the contribution of each term.

Conflicts. The user enters the manuscript’s authors. The tool resolves them to author IDs and flags their co-authors on papers from the last five years, people at the same institution, advisor/student links, and names on the user’s declared conflict list. Flags show the reason. An exclude mode drops them. Details in Appendix F.

Learning weights. The user can rate candidates as good or poor fits. A pairwise logistic regression on the candidates’ features, regularised toward the current settings, learns weights that rank the good fits above the poor ones. The learned weights are saved locally and become the defaults (Appendix G).

5 Evaluation

The evaluation is leave-one-out. Sample 200 indexed papers. For each paper: remove it from the index, use its own vector e_i as the query q , use its OpenAlex reference list as the bibliography, rank all authors, and record where its true authors land. Authors whose only paper is the held-out one cannot be recovered and are excluded. Metrics are mean reciprocal rank of the first true author, and the fraction of true authors in the top 10 and top 20.

index	configuration	MRR	recall@10	recall@20
17 venues, SciNCL	$\lambda=0.3, k=3$	0.189	0.210	0.273
17 venues, SPECTER2	$\lambda=0.3, k=3$	0.193	0.259	0.303
17 venues, SPECTER2	$\lambda=0$ (max pooling)	0.214	0.256	0.320
17 venues, SPECTER2	$\lambda=0.3$ + citation bonus 0.1	0.317	0.324	0.396
29 venues, SPECTER2	$\lambda=0.3$ + citation bonus 0.1	0.324	0.327	0.388

Table 1: Leave-one-out, $n=200$, seed 0. The citation bonus is the largest single gain. The index more than doubled between the last two rows and accuracy held.

This protocol rewards finding the paper’s actual authors among 100,000 candidates. Actual authors are disproportionately prolific, so the preference knobs that reduce the weight of volume, seniority or age lower these numbers by construction (Appendix E). The protocol measures a proxy for reviewer fit, not fit itself. A human-judged evaluation, in the style of Stelmakh et al., would be needed to compare against TPMS-class systems.

A Venues

Operations Research; Management Science; Manufacturing & Service Operations Management; Mathematics of Operations Research; Transportation Science; INFORMS Journal on Computing; Stochastic Systems; Mathematical Programming; SIAM Journal on Optimization; SIAM Journal on Control and Optimization; Production and Operations Management; Journal of Operations Management; Decision Sciences; Naval Research Logistics; IIE Transactions; IIE Transactions; European Journal of Operational Research; Operations Research Letters; Computers & Operations

Research; Annals of Operations Research; Omega; International Journal of Production Economics; International Journal of Production Research; Computers & Industrial Engineering; Transportation Research Part B; Transportation Research Part E; Journal of Optimization Theory and Applications; Journal of the Operational Research Society; Networks.

Add-on collections can be built as separate index directories and searched together with the core: statistics and machine learning (JMLR, JASA, Annals of Statistics, JRSS-B, Biometrika, Annals of Applied Probability; built, 8,621 papers), economics and finance (Journal of Finance, JFE, RFS, Econometrica, AER, QJE, JPE, Marketing Science, JMR), and algorithms (JACM, SIAM J. Computing, Mathematics of Computation). An author who appears only in an add-on is proposed only if they have at least one link to the core literature: a core paper, or a citation to or from one.

B Data provenance

Works are pulled from the OpenAlex REST API with cursor pagination, which has no result cap, filtered on `publication_year` ≥ 2014 and `primary_location.source.id` in the venue list. Works that list a venue only as a non-primary location are included when they have a DOI; this excludes RePEc and repository records misattributed to the journals (361 such records for Management Science alone, 193 of 200 sampled without a DOI). Each pull stores its cursor, counts and filter, and is resumable. The script `check_pull.py` compares stored counts with live counts.

Abstracts are rebuilt from OpenAlex’s inverted index. Missing abstracts are requested from the Semantic Scholar Graph API, 500 DOIs per call. Crossref returns almost none for Elsevier (0 of 25 probed; 17 of 2,262 for other publishers). An Elsevier Article Retrieval pass exists for use with an institutional key.

papers	74,127
with a title (embedded)	73,643
with an abstract	46,302 (62.5%)
with an open-access PDF or landing-page link	23,554 (31.8%)
with a direct PDF link (17-venue index)	18.0%
authorship rows	191,000
distinct authors	101,183
outgoing references (17-venue index)	1.25M from 28,387 papers

Table 2: Index contents on 2026-09-20.

C Side tables

Author statistics. From the OpenAlex authors endpoint, 50 IDs per request: works count, citation count, *h*-index, last known institution.

Editorial boards. Publisher board pages (INFORMS, Elsevier, Wiley, SIAM) refuse scripted access. The tool reads yearly Internet Archive snapshots (2016–2026) of 14 journals’ board pages through archive.org’s public CDX index, parses role headings and names, and stores one row per (journal, role, person) with first and last year seen: 2,667 rows, 1,964 people, 1,323 with a role seen in 2025 or 2026. archive.org rate-limits; requests are spaced 8 s apart.

Genealogy. Advisor and student links from the Mathematics Genealogy Project, fetched by name on request into `data/coi/genealogy.csv`.

Department rosters. 167 OR/IE/OM departments in 25 countries with OpenAlex institution IDs. Rosters were scraped where sites permit: 40 usable, 34 thin, 93 blocked.

D Score terms

Notation: $s_i = \langle q, e_i \rangle$; P_a the papers of author a ; μ_s the mean of s_i over the index; works_a the author’s OpenAlex works count.

$$\begin{aligned} \text{score}(a) = & (1 - \lambda) \max_{i \in P_a} s_i + \lambda \text{mean}_{\text{top-}k}(s_i) + w_{\text{cite}}(1 - 0.5^{n_{\text{cited}}(a)}) + w_{\text{seed}}(\sigma_a - \text{med } \sigma) + w_{\text{early}} \phi_{\text{ec}}(\text{works}_a) \\ & - w_{\text{vol}}(\mathbb{E}[\max \text{ of } |P_a| \text{ random } s] - \mu_s) - w_{\text{ed}} \text{roles}_a - w_{\text{sen}} [\log(1 + \text{works}_a) - \text{med}]_+. \end{aligned}$$

Citation bonus. $n_{\text{cited}}(a)$ is the number of the author’s indexed papers the manuscript cites. The bibliography is parsed from the PDF’s last pages or from pasted text; entries are matched to indexed papers by DOI, then by normalised title. The bonus saturates: 0.5, 0.75, 0.875 of w_{cite} for 1, 2, 3 cited papers.

Seeds. σ_a is the cosine between a ’s profile (mean of e_i over P_a) and the closest seed’s profile. Profile cosines cluster near 1, so the deviation from the pool median is used. Seeds are removed from the output.

Volume correction. With n papers an author has n draws at a high s_i . The expected best-of- n for random papers is computed from the query’s own similarity distribution with a normal approximation, relative to $n = 1$.

Editorial and seniority. A role seen in the last two snapshot years counts 1; an older role counts 0.5. Seniority is log works count above the median author.

Early career. ϕ_{ec} is 1 for one work and falls linearly in log works to 0 at a cutoff (default 15).

Recency (optional). $s_i \leftarrow \mu_s + 0.5^{\text{age}_i/h}(s_i - \mu_s)$. Multiplying the raw cosine does not work: dense-encoder cosines all lie in about $[0.7, 1]$.

Diversity. With parameter d , the top pool is re-ordered by maximal marginal relevance: pick $\arg \max_a (1 - d) \text{score}(a) - d \max_b \cos(a, b)$ over b in the seeds and the candidates already picked.

Computation: a vectorised pass gives each author’s $\max s_i$; the exact loop runs on the top 2,000 (or $100n$) authors plus every cited or seeded author. Penalties only lower scores, so no author outside the pool can outrank the pool. Re-ranking 100,000 authors takes about 40 ms.

E Effect of the preference knobs

Leave-one-out with citation bonus 0.1 on the 17-venue SPECTER2 index (baseline MRR 0.317, recall@10 0.324): volume correction 0.5 gives MRR 0.263 / 0.303; 1.0 gives 0.220 / 0.266; seniority penalty 0.01 gives 0.292 / 0.309; 0.03 gives 0.242 / 0.270; recency half-life 8 years gives 0.070 / 0.087. These knobs express preferences about who should review; they are not accuracy tuning, and the protocol penalises them by construction.

F Conflicts of interest

Manuscript authors are resolved to index authors by normalised name: exact match first, then surname plus initials; ambiguous names are reported. Flagged, with the reason: co-authors on papers within a window (default 5 years); people sharing an institution; advisor/student pairs from the genealogy table; names on the user’s declared list (`~/.kindred/conflicts.txt`), which persists between sessions. Flagging is the default because name resolution is fuzzy; an exclude mode removes flagged candidates. The manuscript’s authors are always removed.

G Learning weights from ratings

- **Features.** Each shown candidate has a feature vector $x_a = (\max s_i, \text{mean}_{\text{top-}k}, \text{cited}, \text{volume}, \text{roles}, \text{seniority})$. The score is $w^\top x_a$, where w is given by the slider settings.
- **Ratings.** The user rates a candidate +1 (good fit) or −1 (poor fit). Unrated candidates shown at the same time are stored as weak negatives with weight 0.3.
- **Pairs.** For each manuscript, every (good, bad) pair gives a difference vector $x_+ - x_-$. A good candidate should score above a bad one, so the target is $w^\top (x_+ - x_-) > 0$.
- **Fit.** Maximise $\sum \log \sigma(w^\top (x_+ - x_-)) - \|w - w_0\|^2$, where w_0 is the current slider setting. The penalty keeps a few ratings from moving the weights far; many ratings move them further. Features are standardised for the fit.
- **Back to sliders.** The text weights are renormalised to sum to 1, so the other weights stay in cosine units. A penalty that comes out with the wrong sign is clipped to 0.
- **Storage.** Ratings go to `~/.kindred/feedback.jsonl`; learned weights go to `~/.kindred/weights.json` and become the slider defaults. Both stay on the machine.
- **Federated option.** The same fit can run across users by averaging their weight vectors. Each user reveals eight numbers and nothing about any manuscript.

H Limitations

37% of papers have no abstract, almost all Elsevier. OpenAlex sometimes splits one author into several IDs or merges several into one; conflict flags inherit this. PhD students appear only if they have a paper in an indexed venue. Many university and publisher sites block scripts, so roster and board data are incomplete. Weights are set by hand or learned from one user’s ratings; there is no shared set of elicited reviewer ratings. Full-text indexing is not attempted: 18% of papers have a direct open-access PDF and INFORMS journals offer no text-mining access.

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